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Master Thesis – Structure Proposal A

Template A: Design, Implementation, Evaluation

by

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About This Proposal

Organising idea. The thesis tells the story of a system: what problem it solves, how it is designed, how it was built on four edges, and how well it works. This is the common structure of computer-science theses and systems papers, and it extends the department template's own *Introduction* and *Related Work* chapters directly. Modelled on general thesis guidance [1, 2] and on systems measurement papers such as MeshInsight [3].

Choose it if WADM itself, the deception layer and its four implementations, is the main contribution, and the benchmark is the means of judging it.

Trade-off. The levelling work and the benchmark-methodology lessons get less space here. They fit in, as Sections 5.6 and 6.6, but they are not the story.

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Page counts are a rough budget, not a department rule. The unibz guidelines advise keeping a chapter under about 20 pages and splitting it above 30 [2].

How to read the rest. The table of contents is the proposed structure. Under each heading is a short note on what to write there, and a *Material* line pointing to the place in the repository that already holds it (`results/` and `plots/` are short for `benchmarks/results/` and `benchmarks/results/plots/`).

Front matter (unchanged from the department template)

- **Title page** and **Eigenständigkeitserklärung**, as generated by `rptuma.cls`. The declaration requires disclosing generative-AI use beyond grammar and style.

- **Abstract** (English): the problem (deception at the web edge has an unknown cost), the approach (one mechanism on four edges, levelled to identical behaviour), the method (paired benchmark on three measurement planes) and the headline numbers.
- **Zusammenfassung**: the same abstract in German.
- **Contents**, **List of Figures**, **List of Tables**.
- **Glossary** (optional, the class's `glossar` environment): WADM, honeytoken, edge, tier, VU, iteration, SLO, cgroup, LuaJIT, PUC Lua, proxy-wasm.

In the department template's own example, websites are cited through an archived copy (`web.archive.org`); do the same for any web source in the list at the end.

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1 Introduction

Why deception at the web edge is worth studying, and what this thesis contributes.

1.1 Motivation

Attackers inspect a web application's HTML, cookies and hidden paths before attacking it. Honeytokens planted there raise high-signal alerts, because no legitimate user touches them. An edge proxy can plant them without changing the application.

1.2 Problem Statement

Nobody has measured what this costs. Which edge technology should host the deception, and what does it cost in latency, CPU, capacity and correctness?

1.3 Research Questions

RQ1: latency added per request. RQ2: CPU and capacity cost. RQ3: whether existing functionality breaks. RQ4: how much the answers depend on the edge's extension runtime.

1.4 Contributions

(i) WADM: one configuration, five honeytoken kinds and a fake SQL-injection trap, implemented on four edges; (ii) behavioural and tuning levelling, with an automated parity check; (iii) a paired, three-plane benchmark harness; (iv) measured costs and where they come from.

1.5 Thesis Outline

One line per chapter.

2 Background

The concepts a reader needs before the design makes sense.

2.1 Cyber Deception, Honeypots and Honeytokens

Spitzner's definition [4, 5]; honeytokens as data with no legitimate use; why false positives are rare.

2.2 Web Application Reconnaissance

What attackers read and tamper with: HTML source, response headers, cookies, `robots.txt`, hidden form fields, SQL-injection probes.

2.3 Reverse Proxies and Their Request Phases

Request hooks, header and body filters, body buffering, upstream connection pools.

2.4 Extension Runtimes

LuaJIT (OpenResty, Envoy Lua), PUC Lua 5.1 (Apache `mod_lua`), WebAssembly through proxy-wasm on V8 (Rust). Interpreted, JIT-compiled and compiled code compared.

Material: `CONTEXT.md` (tech stack table)

2.5 Performance Measurement Fundamentals

Latency percentiles, open- and closed-loop load, coordinated omission, cgroup CPU accounting [6].

3 Related Work

Prior work, grouped by theme, ending with where this thesis sits.

3.1 Honeytokens and Decoys

Decoy documents [7], honeywords [8], and deceptive web artefacts such as HTML comments, `robots.txt` entries and invisible links [9].

3.2 Application-Layer Deception without Code Changes

Kahlhofer and Rass review 19 techniques for deploying deception without developer involvement, reverse proxies among them [10].

3.3 Proxy and Service-Mesh Overhead

Sidecar overhead studies [3]; Lua versus WebAssembly filters in Envoy.

3.4 Benchmarking Methodology in Systems Security

Common benchmarking flaws [11, 12].

3.5 Positioning

No prior work measures deception cost across edge runtimes while holding behaviour identical.

4 Design of WADM

The mechanism on a conceptual level, independent of any one edge.

4.1 Requirements and Threat Model

The operator controls the edge but not the application. The attacker is a reconnaissance or scanning client. Non-goals: blocking traffic, real authentication.

4.2 Architecture

Client → edge → static origin. One shared `config.json`, an in-memory attacker-IP store, and the alert log as WADM's only output.

Material: `CONTEXT.md` (architecture diagram)

4.3 Configuration Model

Per-token `enabled` switch, exact or `/*` paths, and plans precompiled per path when the configuration loads.

Material: `config.json`, `CONTEXT.md` (field tables)

4.4 Honeytoken Kinds

4.4.1 HTML Comments

Injected before `</body>`; detected when the trigger keyword appears in the query.

4.4.2 HTTP Headers

A decoy response header; detected when its value is replayed in a later request.

4.4.3 Cookies

`Set-Cookie` bait; detected on tamper, since a browser replays the value unchanged.

4.4.4 Decoy Paths

Advertised as a hidden link or in `robots.txt`; any request to the path matches.

4.4.5 Form Fields

A hidden `<input>` before `</form>`; detected on tamper.

4.5 The Fake SQL-Injection Trap

The edge answers `POST /api/login` itself: a fake MySQL error that reflects the payload on a signature hit, a canned denial otherwise. Signatures are literal strings, not regular expressions.

4.6 Detection and Response Pipeline

Parse once, scan, strip the parameter so the origin never sees it, record the IP, then write one alert with CR/LF neutralised against log forgery.

4.7 The Canonical Mechanism

Rules every edge must follow: client IP from the socket peer, raw path, ordered query segments, and one plain-find splice primitive.

Material: `docs/EDGE_LEVELING.md` (“Behavioural equivalence and latency pass”)

5 Implementation on Four Edges

How the design maps onto each proxy, and what made the four comparable.

5.1 Common Structure

A table mapping each edge's request hook, header filter and body filter, so the subsections below need only describe the differences.

5.2 OpenResty (Reference Edge)

Configuration compiled once in `init_by_lua`, detection in `access_by_lua`, injection in the header and body filters, attacker IPs in a `lua_shared_dict`.

Material: `nginx/README.md`, `nginx/nginx.conf`

5.3 Envoy with a Lua Filter

`injection.lua`, handing the request path to the response phase via dynamic metadata, body buffering.

Material: `envoy_scripts/README.md`

5.4 Envoy with a WebAssembly Filter

Rust proxy-wasm: configuration compiled in `on_configure`, fewer host calls, a byte-level splice. A proxy-wasm filter cannot answer locally from the body callback.

Material: `wasm-filter/README.md`, `wasm-filter/src/lib.rs`

5.5 Apache httpd with mod_lua

The shared `wadm.lua` core and its three hook scripts; `LuaScope` and `LuaCodeCache`.

Material: `apache_scripts/README.md`

5.6 Levelling the Edges

Product defaults levelled before any comparison: upstream keep-alive, access logs, code caching. Baselines are the WADM configuration minus WADM. The residual differences are listed in a table.

Material: `docs/EDGE_LEVELING.md`, `docs/THESIS_NOTES.md` (2026-09-17, 2026-09-18)

5.7 Verifying Equivalence

`parity_check.py` sends 13 requests to every edge and diffs the responses, the alert lines and the requests that reached the origin.

Material: `benchmarks/parity_check.py`

6 Evaluation Methodology

How the costs were measured, and why the measurements can be trusted.

6.1 Goals and Metrics

Latency added, CPU per request, capacity, share of intact responses, alert recall.

6.2 Testbed

The VM, the cpuset layout, guest tuning, image versions.

Material: `docs/BENCHMARK_METHODOLOGY.md` §4

6.3 Workload

k6 iterations of nine requests: a priming `GET`, four `GET` probes and four SQLi `POST`s. Williams-balanced order [13]; arrival rates of 1, 10 and 100 iterations/s.

Material: `benchmarks/test.js`

6.4 Measurement Planes

In-edge timers (µs), end-to-end latency from k6 (ms), CPU per request from cgroup `cpu.stat`.

6.5 Paired Three-Tier Design

Bare, silent and instrumented tiers run minutes apart in balanced order, over 5 replicates.

Material: `docs/BENCHMARK_METHODOLOGY.md` §7.1

6.6 Run Validity

The `assess_run()` gates, the origin-CPU negative control, and detection of host freezes.

Material: `benchmarks/wadm_timings.py`, `docs/THESIS_NOTES.md` (2026-10-07)

6.7 Statistics and Presentation

Medians of paired differences, the range over replicates, sign agreement, and a 5%-trimmed mean for the timers.

7 Results

What was measured, one aspect per section.

7.1 Latency Added

+109 to +218 µs per `GET` at the median (13–23%). The tail grows more than the median. Apache at 100 iterations/s is an outlier.

Material: `results/paired_headline.md`, `plots/paired_latency_added.png`

7.2 CPU Cost and Its Attribution

CPU added per request: OpenResty +33, Envoy+Lua +41, WASM +123, Apache +200 µs. The in-edge timers enclose only 5–22% of it.

Material: `plots/cost_attribution.png`

7.3 The Cost of Measuring

The timing log lines alone cost about as much as WADM itself on OpenResty and Envoy+Lua.

7.4 Capacity

Share of bare capacity kept with silent WADM: OpenResty 71%, WASM 71%, Envoy+Lua 84%, Apache 30%, each reported with the resource that limited it.

Material: `results/capacity_summary.md`, `plots/capacity.png`

7.5 Per-Mechanism Cost and Page Size

The `GET` path against the trap; injection cost grows linearly with body size.

Material: `plots/mechanism_cpu.png`, `plots/size_sweep.png`

7.6 The SQL-Injection Trap

In the encoding-by-outcome 2×2, percent-decoding dominates scan depth, and only Apache's interpreted Lua makes scan depth visible.

Material: `results/sqli.md`, `docs/THESIS_NOTES.md` (2026-09-21)

7.7 Functional Correctness

Intact responses and alert recall under load. The response-shape matrix: gzip, range requests, HEAD, bodies over 1 MiB, ETags.

Material: `results/paired_functional.md`, `results/functional_matrix.md`

7.8 Stability of Apache mod_lua

Segfaults driven by cumulative requests in the body filter, and the cost of fixing them with `LuaScope conn`.

Material: `docs/BENCHMARK_METHODOLOGY.md` §4b–4c, `plots/apache_remediation.png`

8 Discussion

What the results mean, and how far they can be trusted.

8.1 Choosing an Edge for Deception

A trade-off summary per edge: latency, CPU, capacity, robustness.

8.2 Where the Cost Comes From

Runtime entry and exit, body buffering and log writes, not the detection logic.

8.3 Deployment Implications

Compressed bodies, range requests, preserved ETags, body-size limits.

8.4 Threats to Validity

A single VM host, synthetic traffic, moving image tags, timer resolution.

8.5 Limitations

POST-body inspection exists on only two edges; no real attacker traffic; no TLS or HTTP/2.

9 Conclusion and Future Work

9.1 Summary

One answer per research question.

9.2 Future Work

POST-body inspection on all four edges, injecting into compressed bodies, a field study with real attackers, TLS and HTTP/2.

A Configuration Reference

Every `config.json` field per honeytoken kind, and the SQLi trap's fields.

Material: `CONTEXT.md`

B Parity-Check Request Set

The 13 requests and the documented exclusions.

C Reproducing the Benchmarks

Commands, machine configuration, run durations.

Material: `docs/BENCHMARK_METHODOLOGY.md` §3–4

D Additional Result Tables

Per-rate tables and figures.

Material: `benchmarks/results/plots/by_rate/`

Sources Used for This Proposal

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